

Generalized Kähler Pairs

Kähler MMP

Yi Li

The aim of this note is to introduce the generalized Kähler pairs introduced by [DHY23], which is a transcendental analog of [BZ16].

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1 Generalized b -(1,1)-currents

2 Generalized Singularities

2.1 Conversion between generalized pairs and log pairs

Theorem 2.1 (Generalized klt implies klt, [DHY23, Lemma 2.13]). *Let $(X, B + \beta)$ be a generalized klt (resp. dlt) pair. If $K_X + B$ is $\mathbb{Q}$ -Cartier, then (X, B) is klt (resp. dlt).*

conversely, if the b -(1,1) current descend on X , then klt pair is also a generalized klt pair.

Theorem 2.2.

3 Generalized pairs and Boucksom–Zariski decomposition

4 Applications of the Kähler Generalized Pairs

References

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- [BZ16] C. Birkar and D.-Q. Zhang, *Effectivity of Iitaka fibrations and pluricanonical systems of polarized pairs*, Publ. Math. Inst. Hautes Études Sci. **123** (2016), 283–331.
- [DHY23] O. Das, C.D. Hacon, and J.I. Yáñez, *Mmp for generalized pairs on Kähler 3-folds*, 2023.